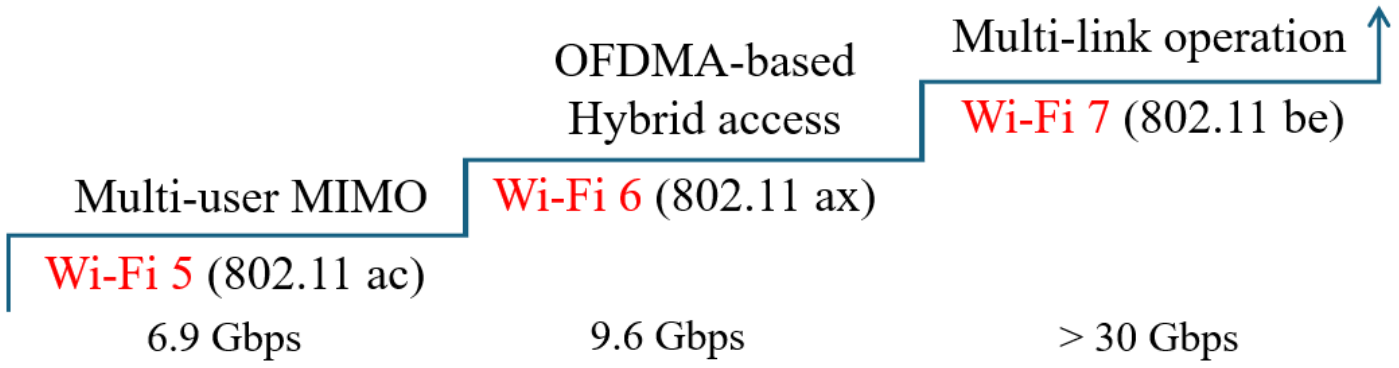
Current and Next-Generation Wi-Fi Protocols: Analysis and Optimization

Arthi S.
Prof. Neelesh B. Mehta.
Dept. of ECE, Indian Institute of Science Bangalore.

Apr. 09, 2026.

Wi-Fi Evolution: New Features, New Challenges

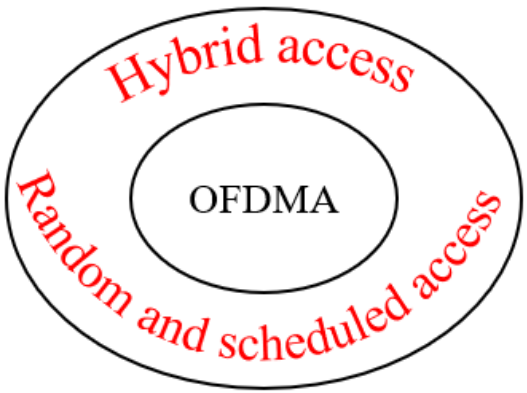
- Wi-Fi connects billions of devices worldwide
  - In homes, enterprises, and even public spaces
  - Significant portion of cellular traffic offloaded to Wi-Fi
- Evolved significantly

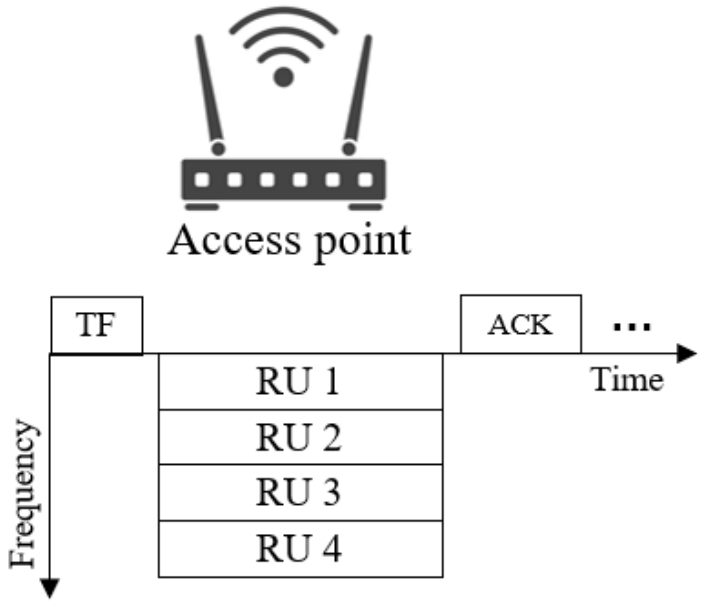


- Effective use requires joint resource allocation and scheduling
- Analysis is essential for explainable performance insights
  - Goal: understand effective use of key features

Outline

- **OFDMA-based hybrid access**
  - Saturation throughput and access delay analysis
  - Dynamic resource allocation design

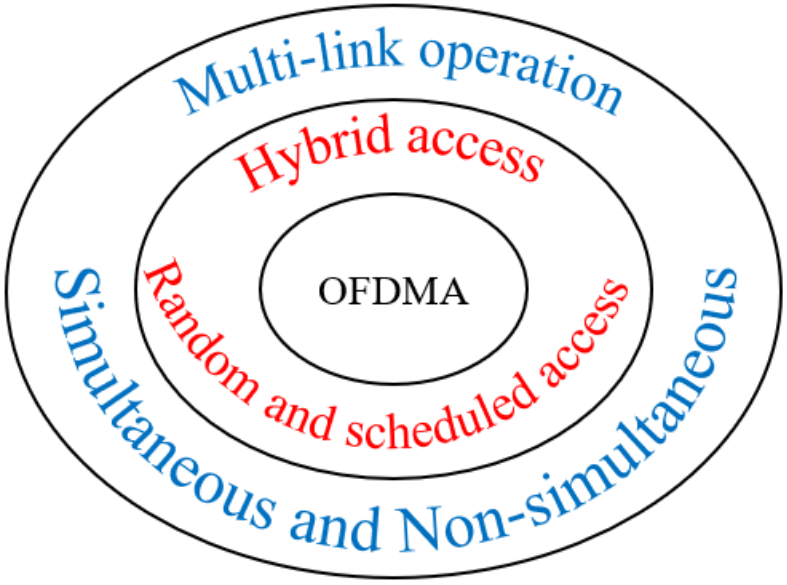




OFDMA in 802.11ax

Outline

- **OFDMA-based hybrid access**
  - Saturation throughput and access delay analysis
  - Dynamic resource allocation design
- **Multi-link operation with hybrid access**
  - Saturation throughput analysis
    - Simultaneous and non-simultaneous multi-link
    - With and without legacy users



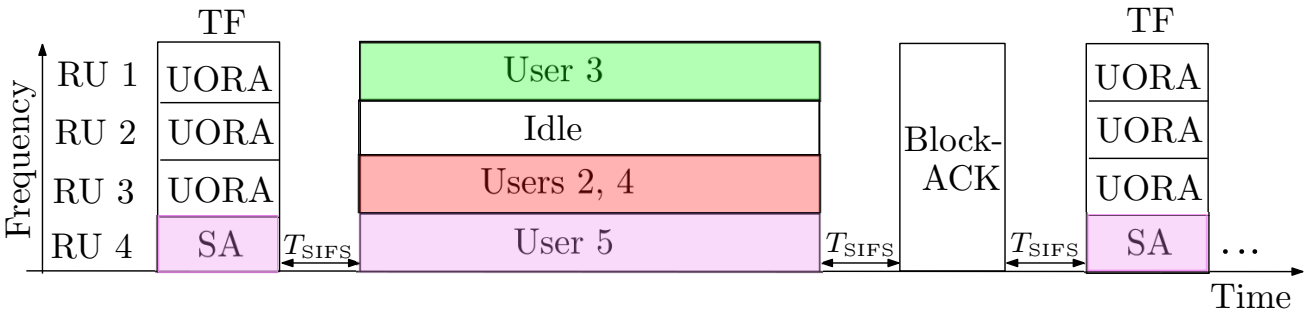


Multi-link in 802.11be

OFDMA-Based Hybrid Access

- AP controls uplink transmissions through trigger frames

User ID	Counter	
	Before TF	After TF
1	7	4
2	1	0
3	2	0
4	3	0



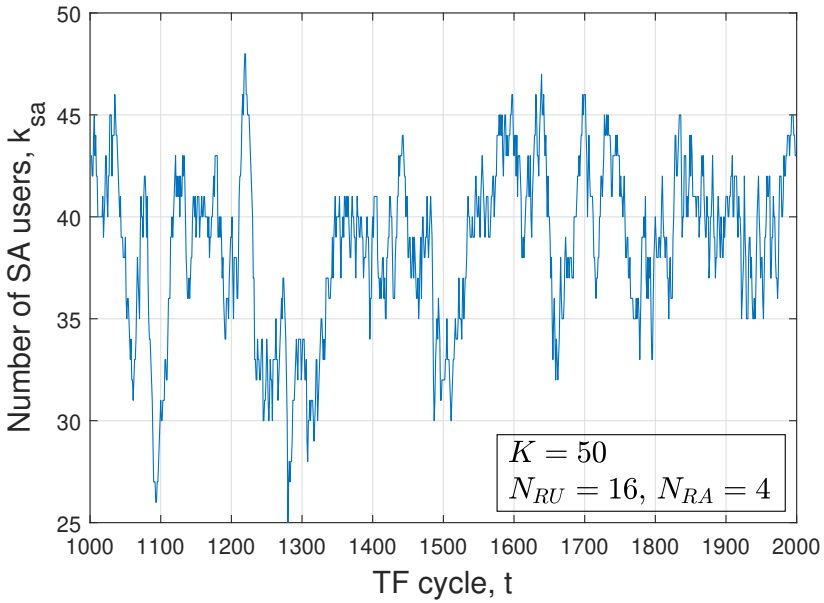
  - Scheduled access: AP schedules users in SA RUs
  - Random access: users contend in UORA RUs to send BSRs

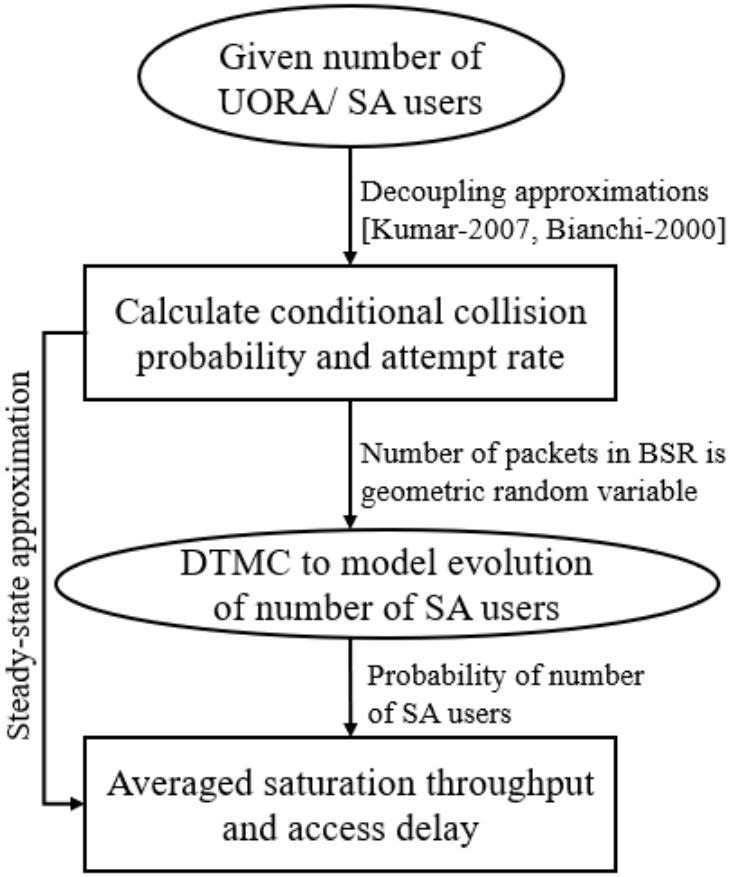
- SA is contention-free and improves throughput
- UORA enables users to enter SA

Looking Under the Hood

Users move between UORA and SA

- Depends on RU allocation, MAC contention, and time user stays in SA
- Modeled with a DTMC

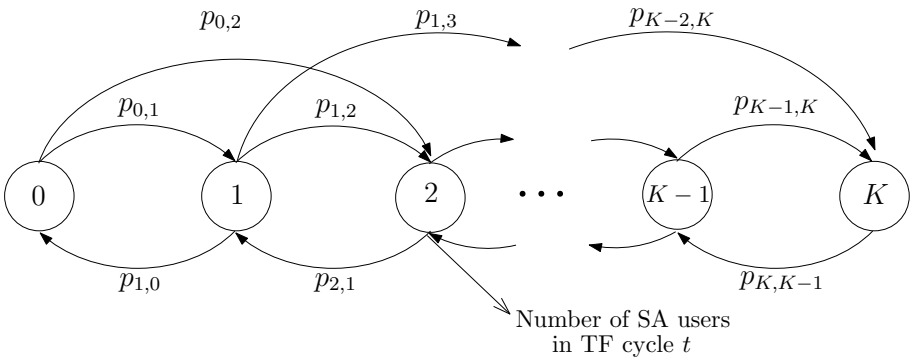




Looking Under the Hood

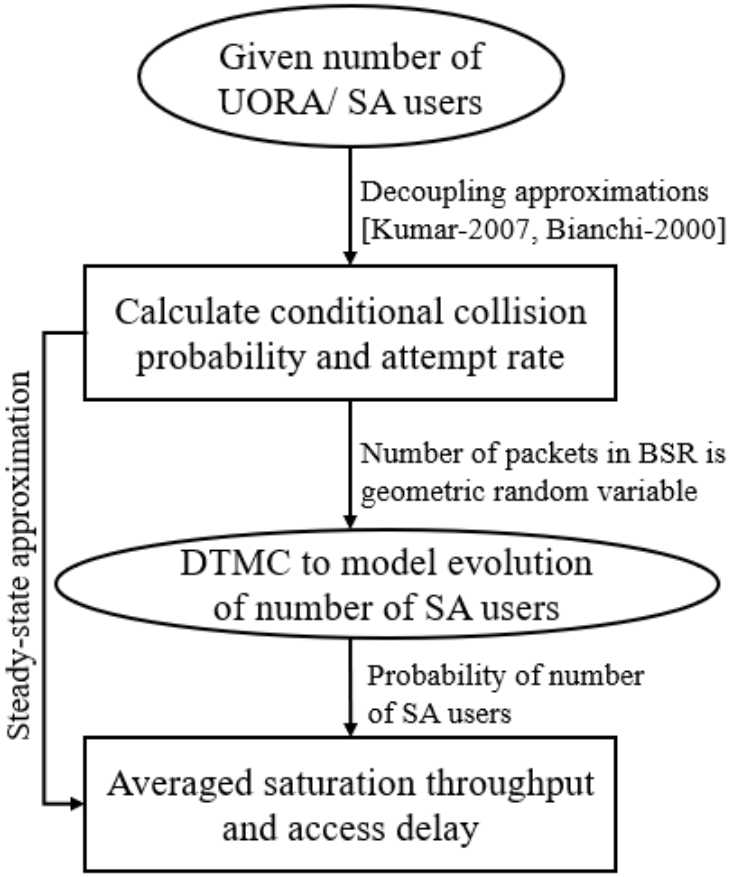
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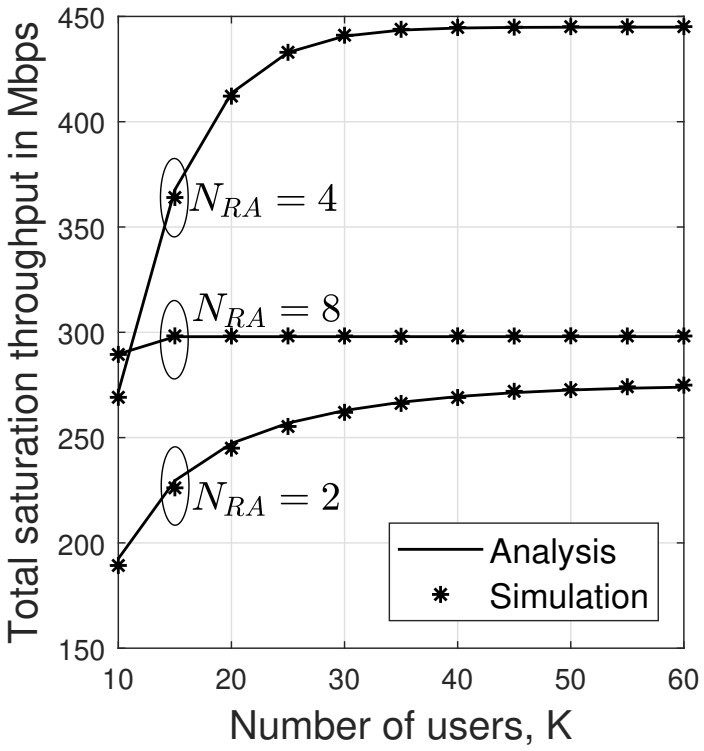


2 UORA RUs and 1 SA RU

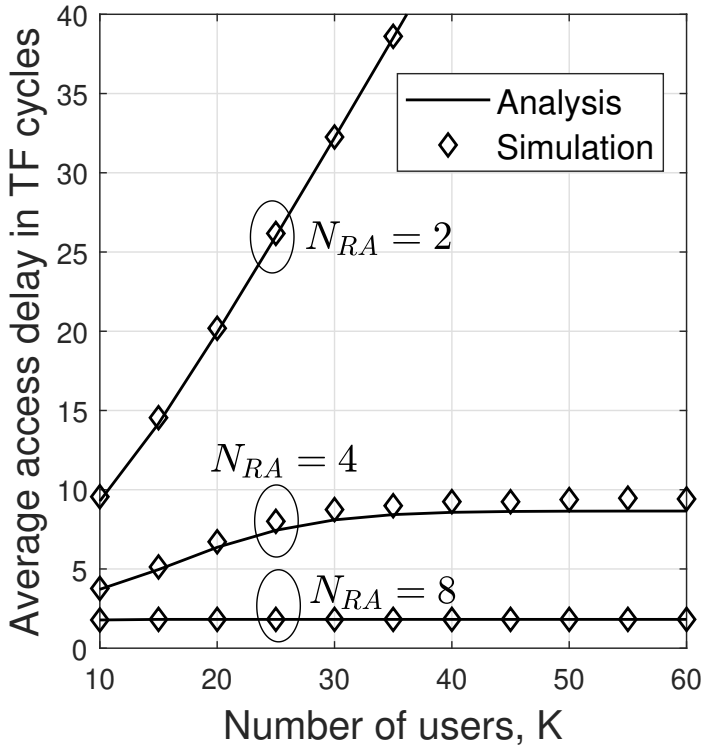
K - Number of users



Saturation Throughput - Access Delay Trade-off



(a) Saturation throughput

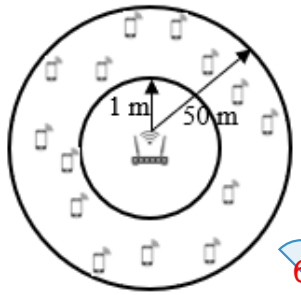


(b) Average access delay

(160 MHz bandwidth, 16 RUs, 5 ms payload duration, and Average number of packets in a BSR = 10)

- Trade-off exists between throughput and access delay

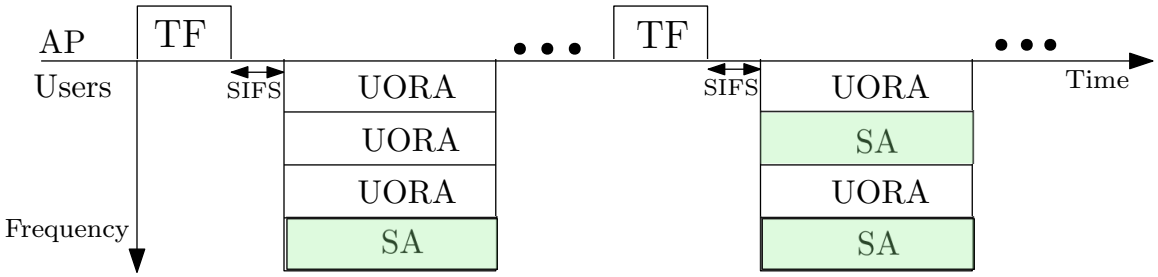
[S. Arthi and N. B. Mehta, IEEE Globecom 2024]

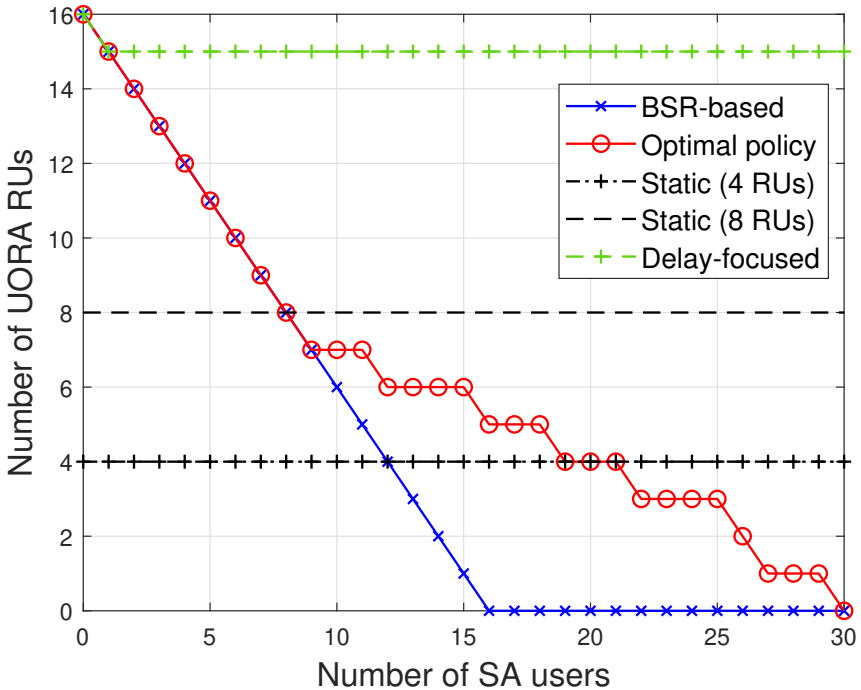


Question: What is the optimal RU allocation policy?

Optimal RU Allocation Policy

- Dynamic RU allocation - RU allocation depends on number of SA users.

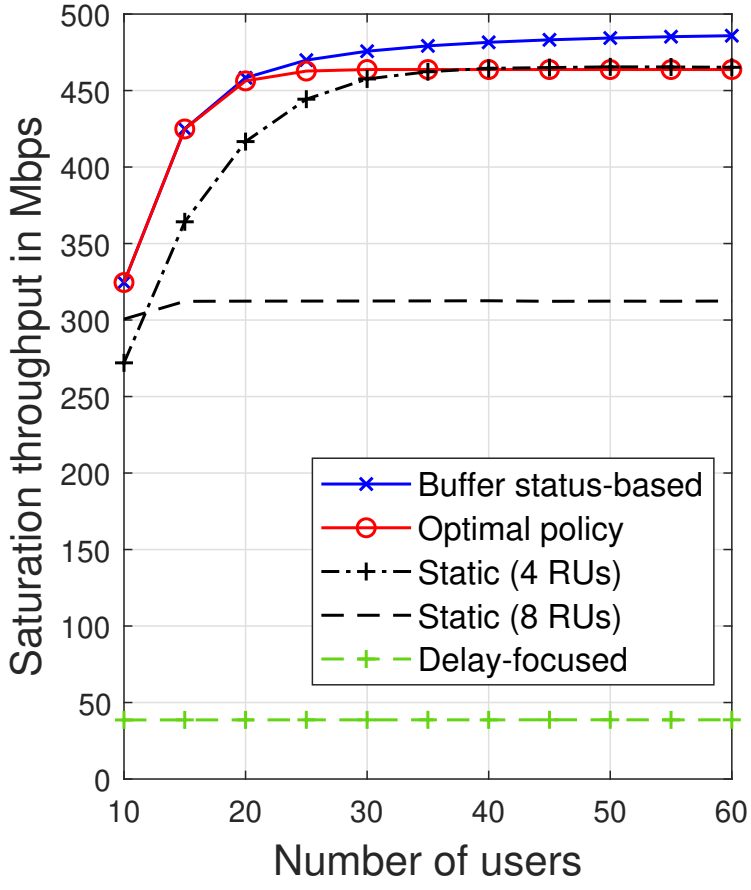




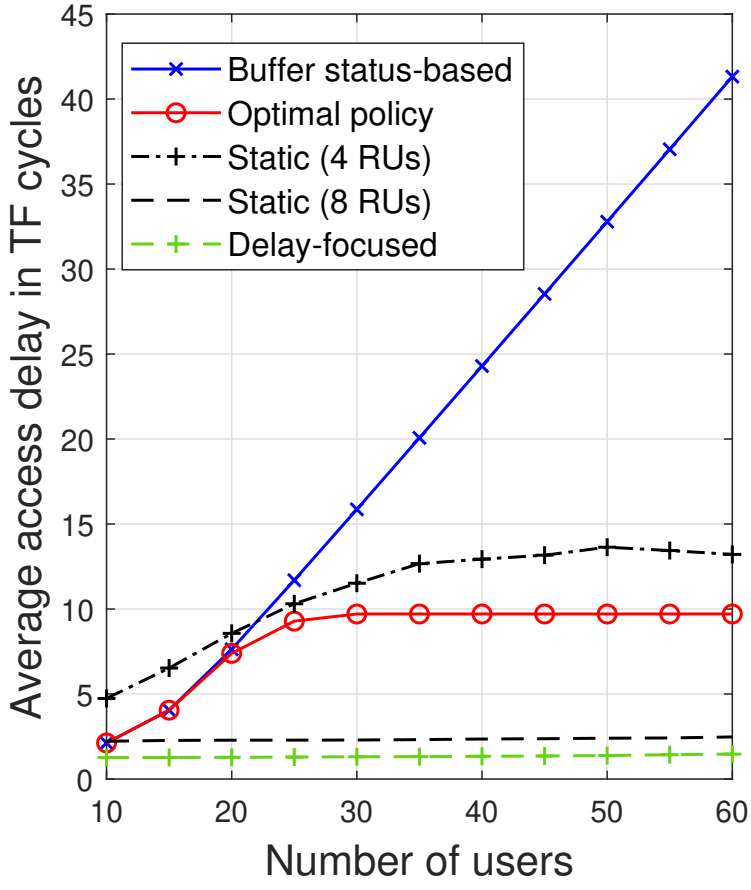
30 **users and** 16 **total RUs**

- Based on the DTMC model
- MDP formulation to find the optimal RU allocation policy
- Optimally balances throughput and delay

Optimal Policy: Saturation Throughput and Access Delay



(a) Saturation throughput



(b) Average access delay

- Our policy optimally balances throughput and access delay

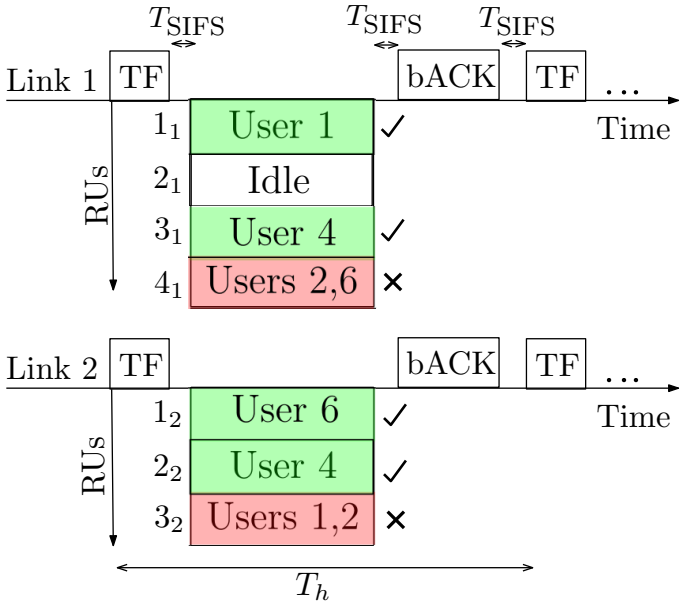
[S. Arthi, N. B. Mehta, and C. Singh, IEEE TMC, Apr. 2026]

Multi-link Operation with Uplink OFDMA Random Access

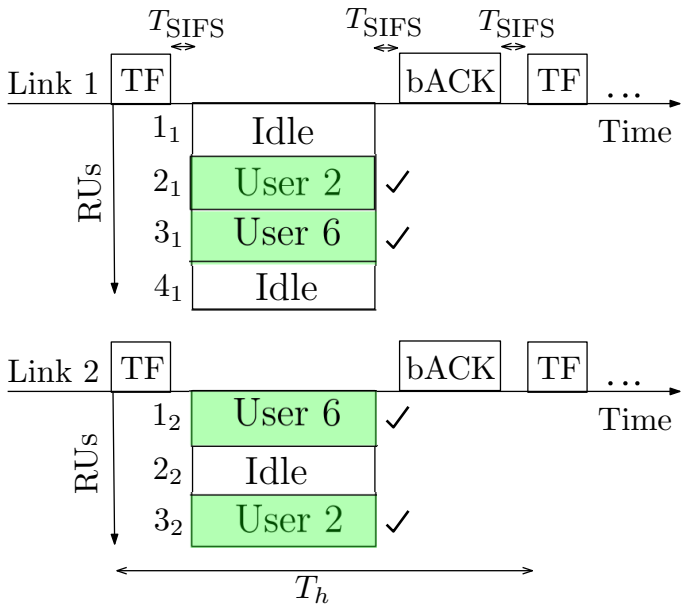
Multi-link + Uplink OFDMA Random Access

- Device can transmit and receive on multiple radio links
- Coordinated back-off mechanisms across links
  - Two possible back-off rules

User ID	Counter	
	Link 1	Link 2
1	0	4
2	0	0
3	2	19
4	31	0
5	1	7
6	0	0

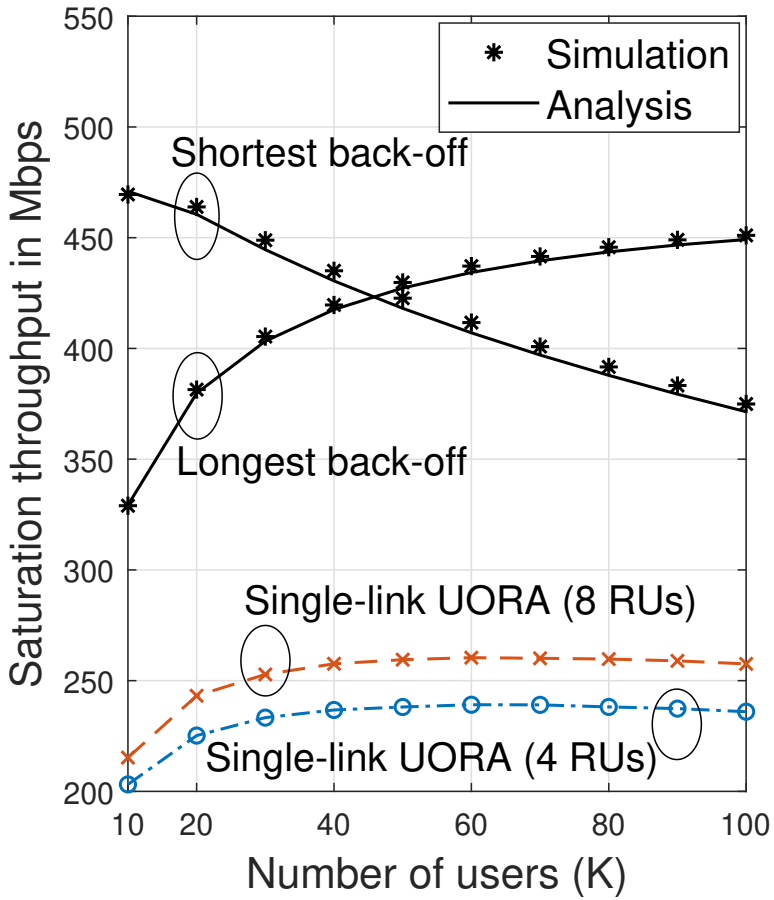


(c) Shortest back-off rule

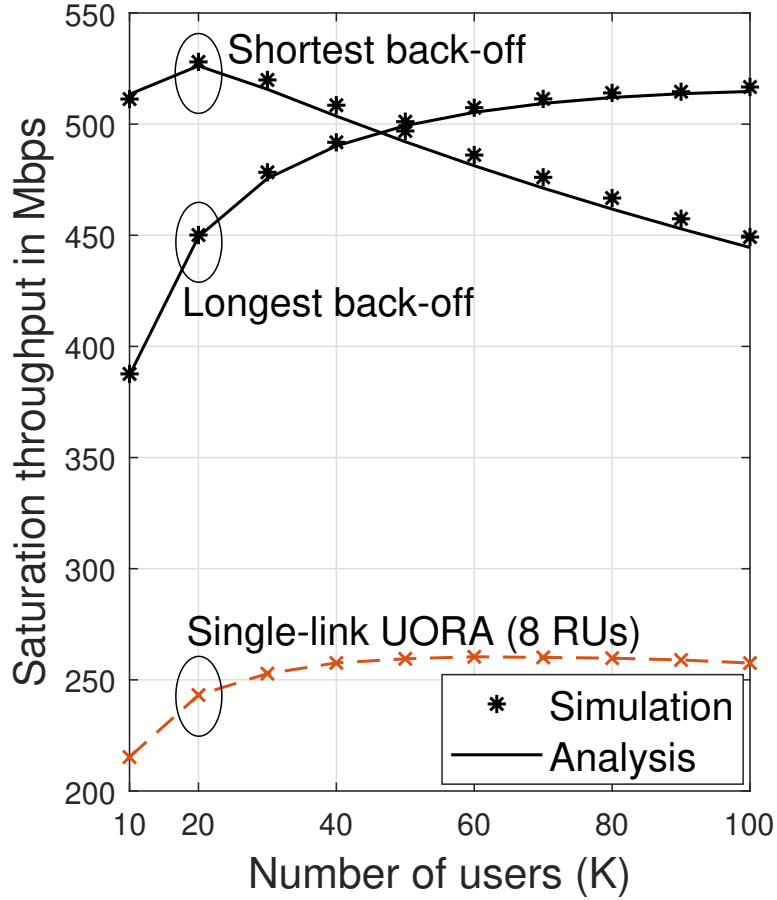


(d) Longest back-off rule

Saturation Throughputs for NSTR MLO + UORA



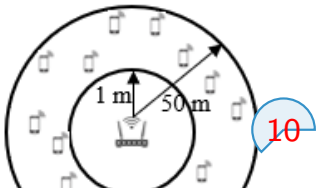
(e) $N_1 = 8$ and $N_2 = 4$



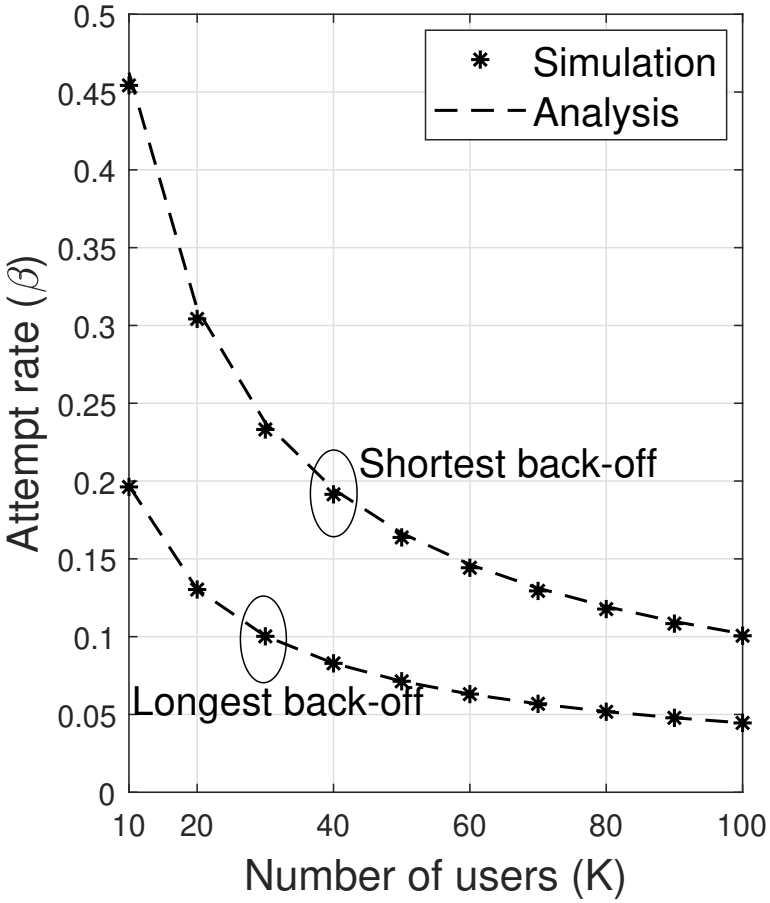
(f) $N_1 = 8$ and $N_2 = 8$

(160 MHz bandwidth, 5 ms payload duration, $CW_{\min} = 16$, $CW_{\max} = 1024$)

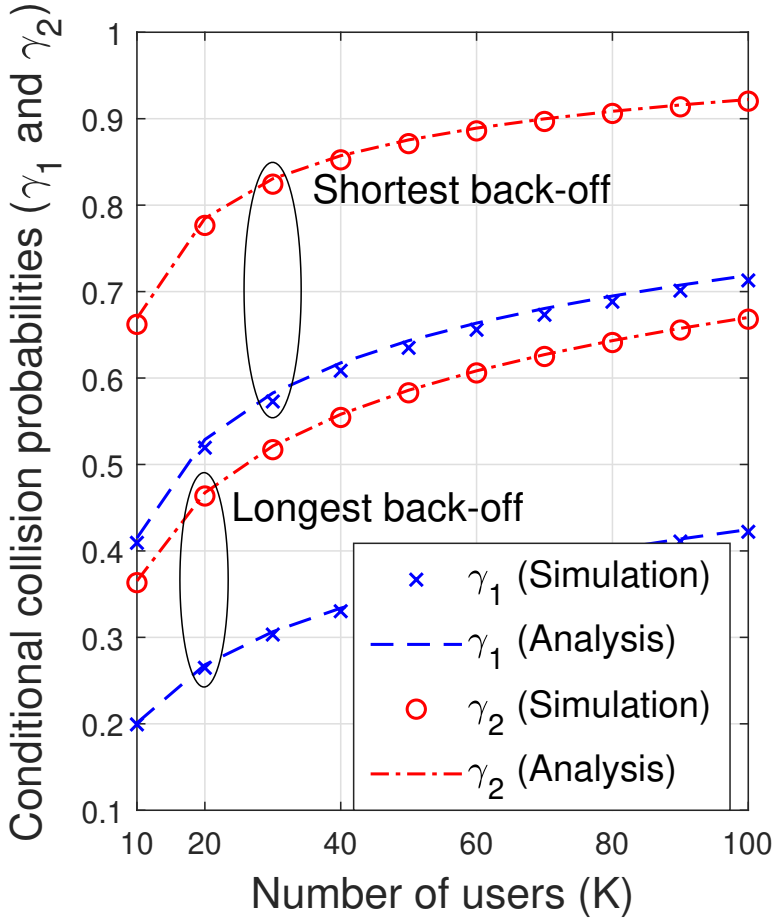
- Shortest back-off has higher throughput for small K
- Longest back-off has higher throughput for larger K



Back-off Parameters when $N_1 = 8$ and $N_2 = 4$



(g) Attempt rate



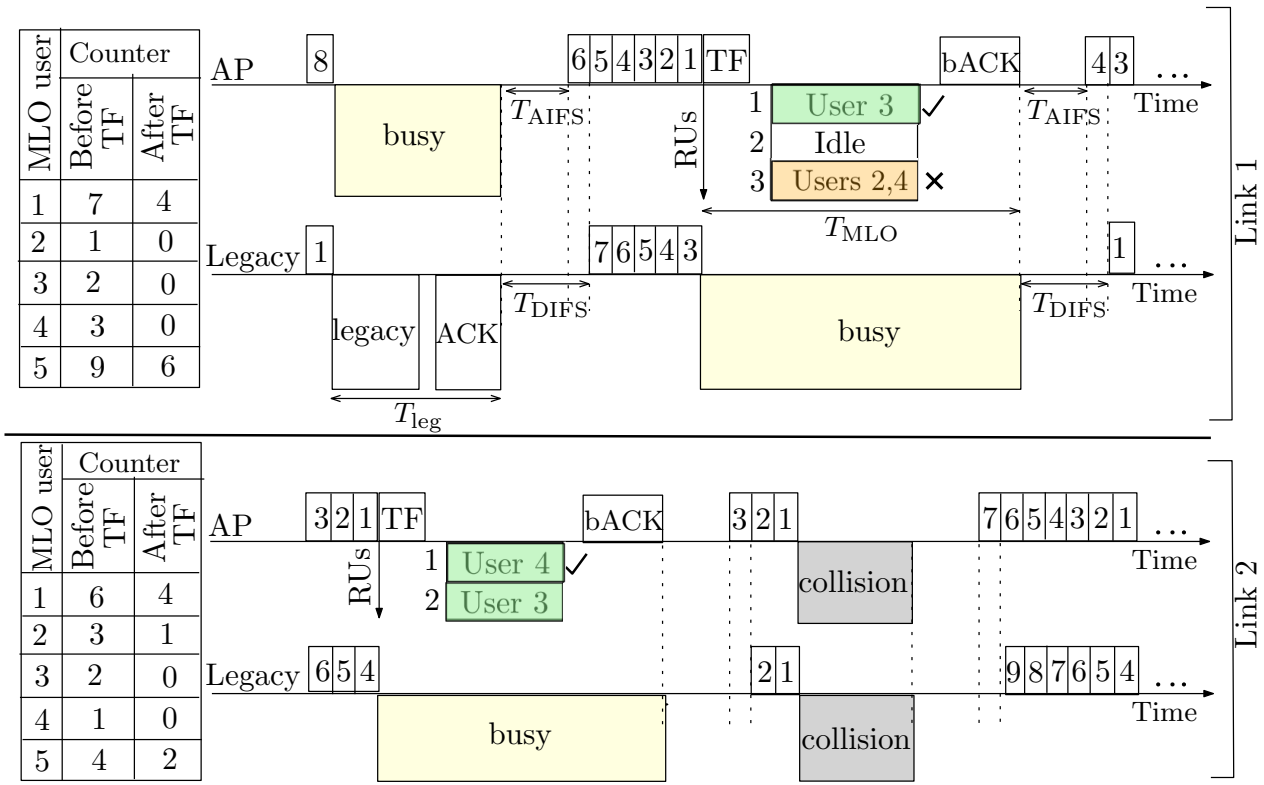
(h) Conditional collision probabilities

- Attempt rate of shortest back-off rule is higher than longest rule
- Conditional collision probability depends on the number of UORA RUs

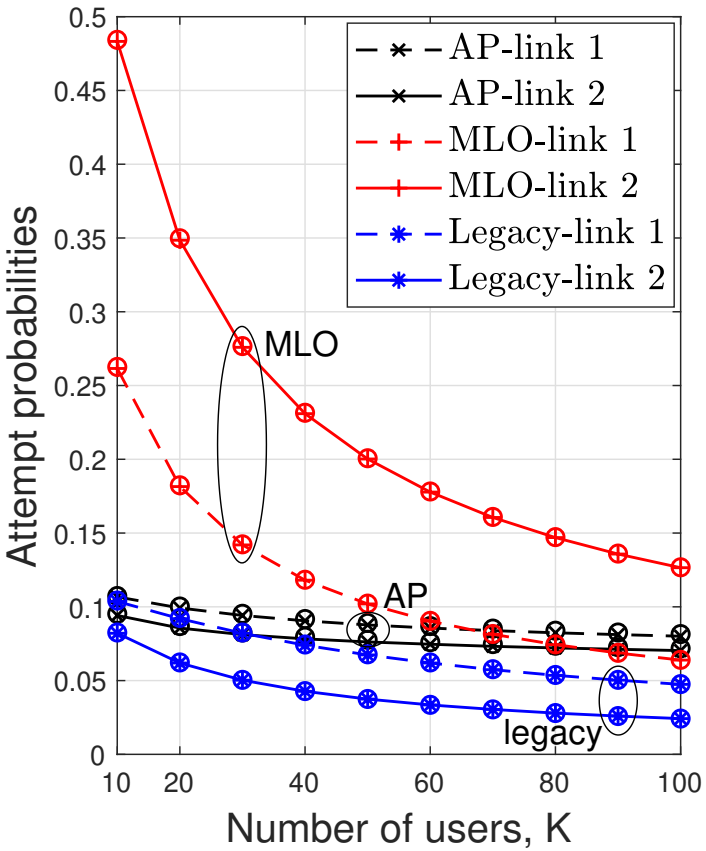
Multi-link + UORA
Now, with Legacy Users

Multi-link + UORA with Legacy users

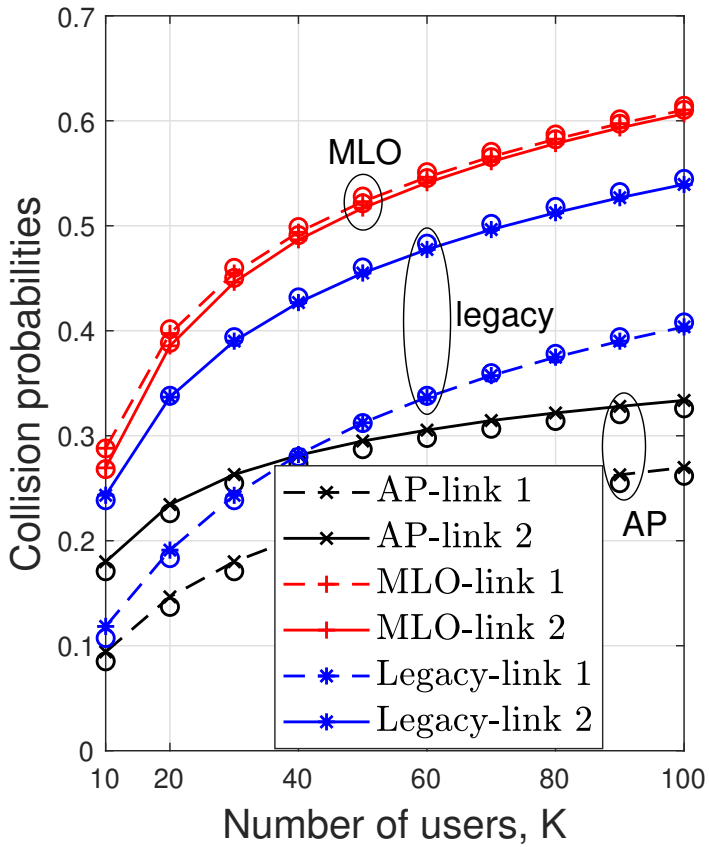
- AP and Wi-Fi 7 users can access both links
  - AP's transmissions are prioritized with EDCA
  - Legacy users must co-exist on each link



Back-off Parameters: MLO + UORA + Legacy Users



(i) Attempt probability

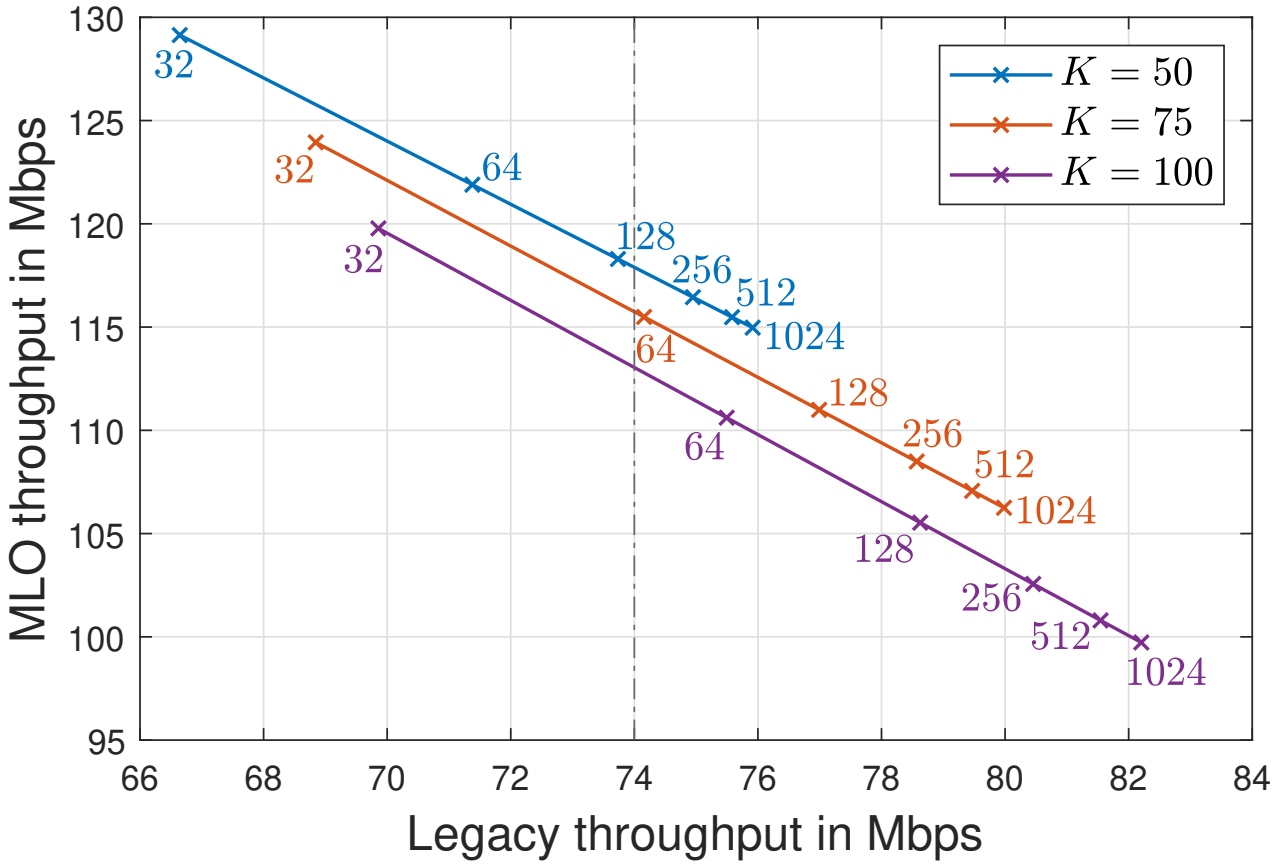


(j) Collision probabilities

($L_1 = 0.1K$, $L_2 = 0.3K$, AIFS $= 3$ slots, DIFS $= 4$ slots,
$C_{\min} = 16$, $C_{\max-AP} = 128$, and $C_{\max-leg} = 1024$)

- AP has higher attempt probability than legacy users

How to Determine Optimal $CW_{\max}$ for AP?



- MLO throughput decreases and legacy throughput increases as $CW_{\max}$ of AP increases

Conclusions

- **Analyzed throughput and access delay of hybrid access**
  - UORA and SA throughputs depended on RU allocation
  - Our policy optimally balanced throughput and delay
- **Analyzed throughput of MLO + UORA**
  - Modeled shortest and longest back-off rules
  - Identified regimes where each back-off rule is preferable
- **Analyzed throughput of MLO + UORA + legacy users**
  - Determined AP's contention parameters
    - For fair co-existence between MLO and legacy users

Publications

- **Hybrid Access OFDMA**
  - S. Arthi and N. B. Mehta, "Saturation throughput analysis of hybrid access MAC protocol in IEEE 802.11ax WLANs," in Proc. IEEE GLOBECOM, Dec. 2024, pp. 1–6.
  - S. Arthi, N. B. Mehta, and C. Singh, "Hybrid Access MAC Protocol in Wi-Fi: Analysis and Optimal Resource Allocation Policy Design," IEEE Trans. Mob. Comput., vol. 25, no. 4, pp. 5742-5757, Apr. 2026.
  - S. Arthi and N. B. Mehta, "Cross-Layer throughput analysis of hybrid access Wi-Fi in time-varying channels," under revision in IEEE Wirel. Comm. Lett., 2026.
- **Multi-link operation**
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Thank you...

Questions?